

Lab 1 Tasks

Advanced Software Development Workshop
2026 Spring

Tasks (before the lab)

(I) Software Engineering (SE) lecture (1 hr 50 min)

- The lab uses material introduced in SE.
- Common mistake students make: try to do the lab without learning the material from SE; get low grade for the lab.

Tasks

(II) Lab (1 hr)

- Explaining the course purpose and rules
- Forming subgroups and teams
- Teamwork issues
- Sign up as subgroup
- Current changes in Software Engineering

Meeting Time Thu 9-11:50

Section Venue	Instructor	TA
1001 T4-301	Judy FENG T3-502-R22 xinfeng@bnnu.edu.cn	Nina MA T3-602-R25 yingranma@bnnu.edu.cn
1002 T29-101	Tianhui MENG T6-403-R7 tianhuimeng@bnnu.edu.cn	Bohan WEN bohanwen@bnnu.edu.cn
1003 T29-201	Jefferson FONG T3-602-R1 jeffersonfong@bnnu.edu.cn	Yitong LIU T3-602-R25 yitongliu@bnnu.edu.cn

Course Purpose

- This workshop allows students to **practice** being a **software engineer** (SE) in **all phases** of **software project development**.
- This includes*
 - Eliciting and documenting specifications
 - Designing software architecture
 - Testing the implementation
 - Communicating as a team
 - Using appropriate tools.

* Explanations are covered in the **SE class**.

Course Purpose

- Advanced Workshop is the companion class to the Software Engineering (SE) course.
 - SE course teaches the theory and principles.
 - In Advanced Workshop , you will put those theory and principles into practice.
 - Through this class, each team will finish one big project, step by step, according to SE principles.

Course Rules

- See the **course documents** in iSpace under the Course Info section:
 - Syllabus
 - CRA (rubric)
 - Workshop Topics and Schedule
- You will work as a **team** for the entire semester.
- You must **submit** your **tasks** on-time, usually 24 hours after class ends, unless otherwise stated in the weekly tasks.
 - We will **not accept late submissions** after the deadline because the graders are on a tight schedule to finish the grading before the next lab.
 - When someone in your team is late in doing his or her part, it's better to submit your team's partial work (get partial credit) rather than not submitting (get no credit).
 - If that person gets his or her part done in the last minutes before the deadline, the leader can resubmit the work before the deadline; the old submission gets overwritten.

Forming Subgroups and Teams

- Students **sign up** as subgroups of **3 people** in iSpace
 - A subgroup can have students enrolled in different sections.
- We randomly combine two subgroups into a team of 6 people.*
 - If all students in a subgroup are from the **same** section, that subgroup will remain in that section.
 - If students in a subgroup are from **different** sections, we will place that subgroup into the section with fewer people to balance out the number of teams in each section; you do not choose which section to belong.
- Students from each team sit together to work during the labs.
 - No need to do add/drop if you physically sit in another section with your team; for enrollment purpose, you are still in the old section.
- Each student should sign up by **today**.

* Some teams might have only 5 people, as determined by the number of students in the class

Teamwork Issues

- Read “Work as a Team.doc” in iSpace
 - T for Talent
 - E for Enthusiasm
 - A for Accountability
 - M for Management
 - W for Work-able
 - O for Openness
 - R for Respect
 - K for Keeness
- There is no “I” in TEAMWORK.

Teamwork Issues

- Teamwork rules:
 - Maintain positive attitude.
 - **Enthusiasm**, the “E” in TEAM.
 - **No whining** – excessive complaints will cause the **team** to **lose points**.
 - Try to resolve whatever problems you have among yourselves.
 - Near the end of the semester, the team **leader evaluates** the team **members**, and team **members evaluate** the team **leader**.
 - **Bonus points** are given to teams (leaders and members) that can resolve their problems within the team.
 - Excellent **individuals ≠ excellent** team.
 - Need to work together.

Teamwork – characteristics of good teamwork

- Worked as a **team, not** as a group of **individuals**.
- Everyone works hard and **respects** each other
 - The “**R**” in TEAMWORK.
 - No one is slacking off.
 - Treat every member as someone useful and important.
- Everyone helps each other
 - If someone doesn’t know how to do the assigned task, other members can help that person.
 - Check other members’ work to see if they were done properly.
- “Everyone gets their own **say**, but **not** everyone gets their own **way**.”
 - You can express your thoughts and feelings, but you cannot demand everything be done your way, especially when someone else in the team disagrees with you.

Teamwork – example 1 of bad teamwork

- The team leader divided the tasks among all the members. **Each individual worked by themselves.**
 - One or two members did their tasks all wrong. They did not attend SE class where we taught how to draw the diagrams.
 - The team got a bad overall grade that week. Other team members complained, **“Why should we have to share their lousy grade?”**
- Their **problem**: they worked as **individuals**, not as a **team**; no “I” in TEAM.
- **Solution**: the members should **check on each other** and **help each other**.
Check your teammates and see if they are doing their tasks properly.
 - Perhaps designate a quality assurance (QA) person to check how everyone is doing every 15 minutes or so during lab. If an individual is having trouble, ask other team members for help, tell him where the material is covered in SE lecture.

Teamwork – example 2 of bad teamwork

- Most of the team worked hard during the lab and even after the lab. One or two members **did very little work, playing** with their **cell phones** during the lab.
- **Problem**: there is a **lack of accountability**; the “A” in TEAM.
- **Solution**: the team leader, with the support of other members, should **warn the slackers**.
 - If they don't change, the leader (with the support of other members) can **give a low grade to the slackers** that week.
 - The QA person (see last example) can check that no one is slacking off during lab.

Teamwork – Characteristics of good team leader

- Characteristics of a **good team leader**
 - Is **not** a **king** or **queen**,
 - **Listens** to other members **rather than dictates** to them,
 - Seeks to build a **consensus** among the team rather than to gives orders,
 - Seeks to make the **team perform its best**,
 - **Manages** people well; assign the **right tasks** to the **right members**,
 - The “**M**” in **TEAM**,
 - Is smart enough, but not necessary the smartest person* in the team.
- If the team leader performs poorly, the team can **vote** for a **new leader**.
 - Shouldn't change leader too often in the semester though.

* In my previous job in a company that create software for airplanes, I have a **PhD**, the team leader has a **master degree** with a lots of domain knowledge. The boss over our teams only has a **bachelor degree**, but he was a former pilot. The big boss could talk to people who use our software. This type of **inverted degree structure** is common in industries. The boss is not the necessary the smartest person.

Teamwork – for discussion or thoughts

- You have participated in many team projects in previous classes. Based on your experiences, answer the following.
 - What are the qualities of a **good team**? Qualities of a **bad team**?
 - What are the qualities of a **good leader**? A **bad leader**?
 - Which person in your team has the qualities of a good leader?
 - How can you make yourself into a good team member?

Sign up as subgroups

- If you haven't done so already, sign up in iSpace as subgroups of 3 people each
- By beginning of Week 2, we will announce the teams in iSpace.
- Starting in Week 2's lab, attend the section venue for your team:
 - Section 1 T4-301
 - Section 2 T29-101
 - Section 3 T29-201
 - All sections meet at Thu 9 – 11:50 am
 - Don't need to do add/drop if you physically sit in another section with your team; for enrollment purpose, you are still in the old section.

Current Changes in Software Engineering

- Currently there is an **ongoing revolution** in Software Engineering, brought on by **AI** and Large Language Models (**LLMs**).
 - Companies, even tech companies, are laying off workers.
 - Software companies such as Microsoft, SAP, Salesforce, CrowdStrike are under pressure.
 - “AI don’t replace workers, workers who use AI will replace workers who don’t.”
Andrew Ng. (See the opening Batch article in iSpace.)
- Unfortunately, we don’t have enough time to change the curriculum of this course to reflect this revolution.

Current Changes in Software Engineering

- However, much of what we teach in this class are still good.
 - First, we've de-emphasized coding, something that AI could do well.
 - Based on my (Jefferson's) decades of experience teaching in colleges and working in companies, fundamental concepts don't become irrelevant; they reappear in different forms.

Current Changes in Software Engineering

- For example, these fundamental questions on SE from Lecture 1 Slide 29-31 are not really addressed in the new AI techniques.

- Questions to answer in **software engineering**
 - **What** is the problem to be solved?
 - **What** are the characteristics of the **software** to solve?
 - **How** will the **software** be constructed?
 - **How** to uncover errors that were made in the design and construction of the **software**?
 - **How** will the **software** be supported **over the long term**, when corrections, adaptations and enhancements are requested by users of the **software**?

- The new AI things can be considered as new tools (Lecture 1 Slide 42) and subdomain (Lecture 1 Slide 38) of SE, rather than something that replace the fundamental concepts of SE that we teach.

My Recommendation to You

- For your own good, you can still learn how AI impacts Software Engineering and CS in general:
 - Read the articles in iSpace; I will put more there throughout this semester.
 - For your FYP, ask your teacher if you can investigate how your teacher's topic or a class that you've taken from him or her is affected by AI:
 - **AI agents** that allow you to automate **processes** using AI tools. (Recall in SE lecture 1, Software Engineering is a process, so it's no surprise that AI has big impact on SE.) The Batch article in iSpace also contains a section on OpenClaw, a new AI agent that's becoming the latest sensation.
 - **LLMs** such as ChatGPT, DeepSeek, Kimi, etc. You are probably familiar with how these can be used to do homework. Bosses also are aware what they can do in the workplace.
 - **Vibe Coding**, a new concept in coding. See the articles in iSpace.